

File permissions in Linux

Project description

Investigating existing permissions on the file system. Determining if the permissions match the authorization that should be given. If they do not match, I'll need to modify the permissions to authorize the appropriate users and remove any unauthorized access. I investigated and updated file and directory permissions for users, groups and others.

Check file and directory details

To check the file and directory details, including the hidden details, the command `ls -la` was used. `ls` lists all the files and directories in that folder, `-l` lists the permissions while the `-a` lists the hidden files, combining them all will give the option `ls -la`.

```
researcher2@cf1133ce6ef30:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Jun 29 09:40 .
drwxr-xr-x 3 researcher2 research_team 4096 Jun 29 10:29 ..
-rw--w---- 1 researcher2 research_team  46 Jun 29 09:40 .project_x.
txt
drwx--x--- 2 researcher2 research_team 4096 Jun 29 09:40 drafts
-rw-rw-rw- 1 researcher2 research_team  46 Jun 29 09:40 project_k.t
xt
-rw-r----- 1 researcher2 research_team  46 Jun 29 09:40 project_m.t
xt
-rw-rw-r-- 1 researcher2 research_team  46 Jun 29 09:40 project_r.t
xt
-rw-rw-r-- 1 researcher2 research_team  46 Jun 29 09:40 project_t.t
xt
```

Describe the permissions string

The permission string is a 10-character string that clearly states the read, write and execute permissions of the user, group and other.

```
researcher2@cf1133ce6ef30:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Jun 29 09:40 .
drwxr-xr-x 3 researcher2 research_team 4096 Jun 29 10:29 ..
-rw--w---- 1 researcher2 research_team  46 Jun 29 09:40 .project_x.
```

Using the first permission string as case study, `drwxr-xr-x`, the string can be divided into four parts: (d) (rwx) (r-x) (r-x). The first part (d) shows that the permission is for a directory, if it was a file it would be (-), second part shows read(r) write(w) execute(x) permission for the user, the third part shows permission for the group and the fourth part shows permission for others.

Change file permissions

To change the file permissions, the `chmod` command is used, this takes 2 parameters, the updated permission and the file name.

```
researcher2@cf1133ce6ef30:~/projects$ chmod o-w project_k.txt
```

Change file permissions on a hidden file

Changing file permissions on a hidden file is similar to the previous step, the only difference is that you must add a (.) before the file name to signify that it is a hidden file.

```
researcher2@cf133ce6ef30:~/projects$ chmod u=r--,g=r--,o=--- .project_x.txt
```

Change directory permissions

The chmod command is still applied while changing directory permissions, as seen in the image below.

```
researcher2@cf133ce6ef30:~/projects$ chmod g-x drafts
```

Summary

Permissions were modified to authorize the appropriate users and remove any unauthorized access. Key permissions guide include: user (u), group(g), others(o), read(r), write(w), execute(x), file(-), directory(d).

```
drwxr-xr-x 3 researcher2 research_team 4096 Jun 29 10:29 ..
-rw--w---- 1 researcher2 research_team  46 Jun 29 09:40 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Jun 29 09:40 drafts
-rw-rw-rw- 1 researcher2 research_team  46 Jun 29 09:40 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Jun 29 09:40 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jun 29 09:40 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jun 29 09:40 project_t.txt
researcher2@cf133ce6ef30:~/projects$ chmod o-w project_k.txt
researcher2@cf133ce6ef30:~/projects$ chmod u=r--,g=r--,o=--- .project_x.txt
researcher2@cf133ce6ef30:~/projects$ chmod g-x drafts
researcher2@cf133ce6ef30:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Jun 29 09:40 .
drwxr-xr-x 3 researcher2 research_team 4096 Jun 29 10:29 ..
-r--r----- 1 researcher2 research_team  46 Jun 29 09:40 .project_x.txt
drwx----- 2 researcher2 research_team 4096 Jun 29 09:40 drafts
-rw-rw-r-- 1 researcher2 research_team  46 Jun 29 09:40 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Jun 29 09:40 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jun 29 09:40 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jun 29 09:40 project_t.txt
researcher2@cf133ce6ef30:~/projects$
```